on a separate device, such as a smartphone. After the user authenticates, they type in the user code and approve the request. The device polls the AS in the background using the device code until the flow completes. If the user approved the request then the device receives an access token the next time it polls the AS.

To initiate the flow, the device first makes a POST request to a new device authorization endpoint at the AS, indicating the scope of the access token it requires and authenticating using its client credentials. The AS returns three details in the response:

- A *device code*, which is a bit like an authorization code from chapter 7 and will eventually be exchanged for an access token after the user authorizes the request. This is typically an unguessable random string.
- A *user code*, which is a shorter code designed to be manually entered by the user when they approve the authorization request.
- A *verification URI* where the user should go to type in the user code to approve the request. This will typically be a short URI if the user will have to manually type it in on another device.

Listing 13.16 shows how to begin a device grant authorization request from Java. In this example, the device is a public client and so you only need to supply the `client_id` and `scope` parameters on the request. If your device is a confidential client, then you would also need to supply client credentials using HTTP Basic authentication or another